

MIAE 221: Materials Science for Engineers

Chapter 9: Phase Diagrams & Microstructural Evolution (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

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(Syllabus Week 8 · Post-Midterm Pillar & Heavily Tested on Final Exam)

1. 🎯 30-Second Core Intuition & Plain-English Concept

A phase diagram is a metallurgical roadmap showing what microscopic phases exist at any combination of temperature and chemical composition.

From a phase diagram, you can answer three vital questions for any alloy:

1. **What phases are present?** (Look at which field the point lands in).
2. **What is the chemical composition of each phase?** (Draw a horizontal **tie-line** and read the intersections).
3. **How much of each phase is present?** (Apply the **Inverse Lever Rule**).

Real-World Analogy

: Making chocolate milk. Below the solubility limit, cocoa dissolves completely into milk (single liquid phase). Add too much cocoa powder, and solid sludge settles at the bottom: now you have two phases (saturated liquid milk + solid cocoa powder) in equilibrium.

2. ⚙️ High-Yield Mathematical Engine & The Lever Rule

1. Gibbs Phase Rule (Condensed System at 1extatm)

$$P + F = C + 1$$

- P : Number of phases present.
- F : Degrees of freedom (number of externally controllable variables: T , composition).
- C : Number of chemical components (e.g., $C = 2$ for binary systems like Cu-Ni or Fe-C).

2. The Inverse Lever Rule (Phase Fraction Calculation)

In a two-phase region ($\alpha + L$) with overall alloy composition C_0 :

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L} \quad (\text{Fraction of Liquid})$$

$$W_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L} \quad (\text{Fraction of Solid } \alpha)$$

(Notice the inverse nature: the amount of α is proportional to the lever arm on the liquid side!)

3. Core Invariant Reactions

- **Eutectic:** $L \xrightarrow{\text{cooling}} \alpha + \beta$ (Liquid freezes into two intimate solid phases).
- **Eutectoid:** $\gamma \xrightarrow{\text{cooling}} \alpha + \beta$ (Solid transforms into two new solid phases).
- **Peritectic:** $L + \alpha \xrightarrow{\text{cooling}} \beta$.

4. The Iron-Carbon ($Fe - Fe_3C$) System (The Heart of Metallurgy)

- **Ferrite (α):** BCC iron. Stable at room temperature. Extremely low carbon solubility (max 0.022 wt at 727°C). Soft and ductile.
- **Austenite (γ):** FCC iron. Stable between 912°C and 1394°C. High carbon solubility (max 2.14 wt at 1147°C).
- **Cementite (Fe_3C):** Stoichiometric iron carbide containing 6.70 wt % Carbon. Extremely hard and brittle.
- **Eutectoid Reaction at 727°C and 0.76 wt %:**

$$\gamma(0.76)$$

Microstructure

: Alternating microscopic lamellae (plates) of soft ferrite and hard cementite.

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

with binary alloys—those that contain two components. If more than two components are present, phase diagrams become extremely complicated and difficult to represent. An explanation of the principles governing and the interpretation of phase diagrams can be demonstrated using binary alloys even though most alloys contain more than two components.

Binary phase diagrams are maps that represent the relationships between temperature and the compositions and quantities of phases at equilibrium, which influence the microstructure of an alloy. Many microstructures develop from *phase transformations*, the changes that occur when the temperature is altered (typically upon cooling). This may involve the transition from one phase to another or the appearance or disappearance of a phase. Binary phase diagrams are helpful in predicting phase transformations and the resulting microstructures, which may have equilibrium or nonequilibrium character.

ISOMORPHOUS SYSTEMS

Possibly the easiest type of binary phase diagram to understand and interpret is the type that is characterized by the copper–nickel system (Figure 9.3a). Temperature is plotted along the ordinate, and the abscissa represents the composition of the alloy, in weight percent of nickel. The composition ranges from 0 wt% Ni (100 wt% Cu) on the far left horizontal extreme to 100 wt% Ni (0 wt% Cu) on the right. Three different phase regions, or *fields*, appear on the diagram: an alpha (α) field, a liquid (L) field, and a two-phase $\alpha + L$ field. Each region is defined by the phase or phases that exist over the range of temperatures and compositions delineated by the phase boundary lines.

Figure 9.3: Tie-line and lever rule construction in a binary isomorphous system.

9.18 THE IRON-IRON CARBIDE ($\text{Fe}-\text{Fe}_3\text{C}$) PHASE DIAGRAM

ferrite
austenite

A portion of the iron–carbon phase diagram is presented in Figure 9.24. Pure iron, upon heating, experiences two changes in crystal structure before it melts. At room temperature, the stable form, called **ferrite**, or α -iron, has a BCC crystal structure. Ferrite experiences a polymorphic transformation to FCC **austenite**, or γ -iron, at 912°C (1674°F). This austenite persists to 1394°C (2541°F), at which temperature the FCC austenite reverts back to a BCC phase known as δ -ferrite, which finally melts

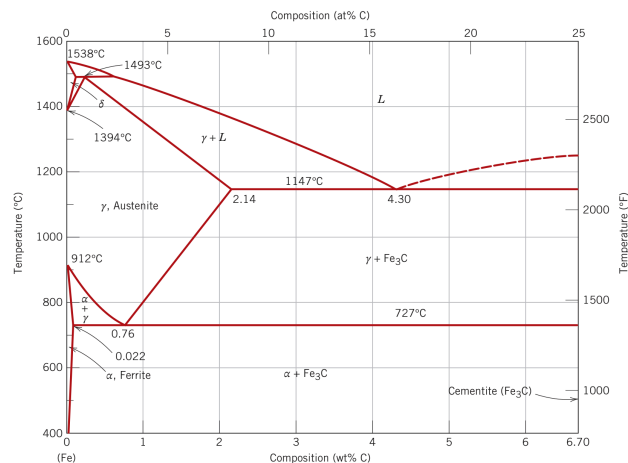


Figure 9.24 The iron–iron carbide phase diagram.
[Adapted from *Binary Alloy Phase Diagrams*, Vol. 1, 2nd edition, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

Figure 9.24: The Iron-Iron Carbide ($\text{Fe} - \text{Fe}_3\text{C}$) phase diagram showing ferrite (α), austenite (γ), and cementite (Fe_3C).

In-Depth Visual Breakdown:

- **The Lever Rule (Figure 9.3):** At point B (temperature T_0 , composition C_0):
 - Draw horizontal tie-line from solidus line C_{α} to liquidus line C_L .
 - The total tie-line length is $(C_{\alpha} - C_L)$.
 - The weight fraction of liquid W_L is the length of the segment opposite to the liquidus line, divided by total length: $W_L = \frac{C_{\alpha} - C_0}{C_{\alpha} - C_L}$.
- **The $Fe - Fe_3C$ Phase Diagram (Figure 9.24):** The cornerstone of mechanical engineering materials:
 - **Steels** contain < 2.14 wt (typically $0.05 - 1.2$ wt).
 - **Cast Irons** contain > 2.14 wt (typically $3.0 - 4.5$ wt).
 - **Hypoeutectoid Steels** ($C_0 < 0.76$): Cool to form proeutectoid ferrite + pearlite.
 - **Hypereutectoid Steels** ($C_0 > 0.76$): Cool to form proeutectoid cementite + pearlite.

4. Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Calculating the fraction of **proeutectoid ferrite** vs. **eutectoid ferrite** vs. **total ferrite** in hypoeutectoid steels.
- **The Classic Concordia Exam Trap:**
 - **Total Ferrite vs. Proeutectoid Ferrite:**
 - **Total Ferrite (α_{total})** is calculated across the entire baseline tie-line at 726°C (0.022 to 6.70):

$$W_{\alpha, \text{total}} = \frac{6.70 - C_0}{6.70 - 0.022}$$

- **Proeutectoid Ferrite (α_{pro})** is calculated
just above the eutectoid temperature

(728°C) between 0.022 and 0.76:

$$W_{\alpha, \text{pro}} = \frac{0.76 - C_0}{0.76 - 0.022}$$

- **Eutectoid Ferrite** is the ferrite residing
inside

the pearlite colonies:

$$W_{\alpha, \text{eutectoid}} = W_{\alpha, \text{total}} - W_{\alpha, \text{pro}}$$

5. Master Exam Problem & Step-by-Step Solution Framework

Problem (Classic Concordia Final Exam Question):

For a 0.35 wt plain carbon steel cooled slowly to just below 727°C , calculate: (a) The mass fraction of total ferrite (α) and total cementite (Fe_3C), and (b) The mass fraction of proeutectoid ferrite and pearlite.

- **Step 1: Compute Total Phases Just Below 727°extC (Tie-line: 0.022 to 6.70)**

$$W_{\alpha,\text{total}} = \frac{6.70 - 0.35}{6.70 - 0.022} = \frac{6.35}{6.678} = 0.951 \quad (95.1$$

$$W_{Fe_3C,\text{total}} = \frac{0.35 - 0.022}{6.70 - 0.022} = \frac{0.328}{6.678} = 0.049 \quad (4.9$$

- **Step 2: Compute Microconstituents Just Above 727°extC (Tie-line: 0.022 to 0.76)**

At 728°extC , the steel consists of proeutectoid α and austenite γ

:

$$W_{\alpha,\text{pro}} = \frac{0.76 - 0.35}{0.76 - 0.022} = \frac{0.41}{0.738} = 0.556 \quad (55.6$$

$$W_{\gamma} = \frac{0.35 - 0.022}{0.76 - 0.022} = \frac{0.328}{0.738} = 0.444 \quad (44.4$$

- **Step 3: Relate Austenite to Pearlite**

Upon cooling through 727°extC , all remaining austenite of composition 0.76 transforms 1-to-1 into pearlite

:

$$W_{\text{pearlite}} = W_{\gamma} = 0.444 \quad (44.4$$

- **Step 4: Verify Consistency**

$$W_{\alpha,\text{pro}} + W_{\text{pearlite}} = 0.556 + 0.444 = 1.000 \quad (100$$

Amount of eutectoid ferrite inside pearlite

:

$$W_{\alpha,\text{eutectoid}} = 0.951 - 0.556 = 0.395 \quad (39.5$$
